LESSON 6: SPECTRAL THEOREM FOR BOUNDED OPERATORS ON A HILBERT SPACE

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In this lecture, we will consider the following fact: the spectral theorem in its functional calculus form. This theorem is based on the Riesz-Markov representation theorem, which is the theorem of representation of linear functionals on the space of continuous functions. This representation theorem uses two notions: the Stieltjes integral and functions of bounded variation.

In the first part of the lecture, we will provide an overview of the Stieltjes integral within the space of functions of bounded variation and discuss the Riesz-Markov theorem. Once this theorem is well understood, we will extend the continuous functional calculus to bounded and measurable functions. This is the spectral theorem in the functional calculus form.

The main discussions regarding the facts that led to the Riesz-Markov representation theorem are included in an appendix at the end of the notes (In order to familiarize ourselves with these results, we will provide more details). At the beginning of the notes, we summarize the discussion of the appendix along with further considerations that are preliminary to the spectral theorem.

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1. Summary of the results we need for the spectral theorem

We summarize what we need for the Spectral theorem. For more details see the appendix A.

Definition 1.1 (Riemann-Stieltjes)**.** *We recall the definition and the main properties of the Stieltjes Integral* $\int_a^b f(t)dF(t)$ *on a bounded interval* $[a,b]$*.*

Assume $F : [a,b] \to \mathbb{R}$ *is right-continuous and increasing.*

Assume $f : [a,b] \to \mathbb{C}$ *is continuous.*

Then the integral is defined in the usual way as the limit of the Riemann-Stieltjes sums

$$\lim_{\delta \to 0} \sum_{i=0}^{n} f(\xi_i)\left(F(t_i) - F(t_{i-1})\right)$$

where $a = t_0 < t_1 < \cdots < t_n = b$*,* $\xi_i \in [t_{i-1}, t_i]$*, and* $\delta = \max_i(t_i - t_{i-1})$*.*

Remark 1.2. *If it turns out that* $F \in C^1([a,b])$*, then we have*

$$\int_a^b f(t)dF(t) = \int_a^b f(t)F'(t)dt.$$

As it is possible to extend the Riemann integral to the Lebesgue integral, the Riemann-Stieltjes integral can also be extended to the Lebesgue-Stieltjes integral. We will not provide details but will just mention some facts.

Definition 1.3 (Stieltjes Integral for Bounded and Measurable Complex-Valued Functions)**.** *We will need to integrate function* $f : [a,b] \to \mathbb{C}$ *that is bounded and measurable*

and we denote this class by $B([a,b])$. We consider the case of simple functions

$$f_n(t) = \sum_{j=1}^{n} \alpha_j \chi_{I_j}(t)$$

where for $a = t_0 < t_1 < \cdots < t_n = b$ and the functions $\chi_{I_j}(t) = 1$ if $t \in [t_{i-1}, t_i)$ and zero otherwise and $f(b) = \alpha_n$. We then define:

$$\int_a^b f(t)dF(t) = \lim_{n\to\infty} \int_a^b f_n(t)dF(t),$$

where $F(b^-) = \lim_{t\to b^-} F(t)$. It can be verified that this limit exists and does not depend on the sequence f_n, but only on f.

Definition 1.4 (Definition of functions of bounded variation complex valued)**.** *We say that a function $F : [a,b] \to \mathbb{C}$ is a bounded variation if F can be written in the form:*

$$F(t) = F_1(t) - F_2(t) + i\left(F_3(t) - F_4(t)\right),$$

where F_i are

- *real,*
- *increasing,*
- *continuous on the right.*

Remark 1.5. *This is the definition that we use for functions of bounded variation. It is based on the Jordan decomposition and we remove the ambiguity by adding continuity on the right.*

Now we define the integral $\int_a^b f(t)dF(t)$ for $F : [a,b] \to \mathbb{C}$ of bounded variation.

Definition 1.6 (Integral with respect to a functions of bounded variation)**.** *Let $f : [a,b] \to \mathbb{C}$ be bounded and measurable, i.e. $f \in B([a,b])$. Let $F : [a,b] \to \mathbb{C}$ be a function of bounded variation. We define*

$$\begin{aligned}\int_a^b f(t)dF(t) :&= \int_a^b f(t)dF_1(t) - \int_a^b f(t)dF_2(t) \\ &\quad + i\left(\int_a^b f(t)dF_3(t) - \int_a^b f(t)dF_4(t)\right),\end{aligned}$$

Remark 1.7. *The integrals have the usual properties of linearity*

Theorem 1.8 (Riesz-Markov Representation Theorem)**.** *Let $C([a,b])$ denote the space of continuous functions on the closed interval $[a,b]$ with the supremum norm:*

$$\|f\|_\infty = \sup_{t\in[a,b]} |f(t)|.$$

For every bounded linear functional Λ on $C([a,b])$, there exists a unique function F of bounded variation on $[a,b]$ such that

$$\Lambda(f) = \int_a^b f(t)\, dF(t), \quad \forall f \in C([a,b]).$$

Furthermore, the norm of the functional Λ is given by the total variation of F, i.e.,

$$\|\Lambda\| = \|F\|_{TV},$$

where $\|F\|_{TV}$ denotes the total variation norm of F.

Consequences of the Riesz-Markov Representation Theorem:

Theorem 1.9 (Uniqueness of the function F)**.** *Let F_1 and F_2 be functions of bounded variation such that*

$$\int_a^b f(t)\, dF_1(t) = \int_a^b f(t)\, dF_2(t), \quad \forall f \in C([a,b]).$$

Then $F_1(t) = F_2(t), \quad t \in [a,b]$. This fact is a consequence of the uniqueness if the function F as stated in the representation theorem.

Remark 1.10 (Extension)**.** *Let F be of bounded variation then from the representation theorem*

$$\Big|\Lambda(f)\Big| = \Big|\int_a^b f(t)\, dF(t)\Big| \leq \|\Lambda\| \sup_{t\in[a,b]} |f(t)|, \quad \forall f \in C([a,b]).$$

It is clear that the above estimate still holds for the larger class of functions $f \in B([a,b])$.

2. Spectral Theorem as functional calculus (version 1)

Remark 2.1 (Considerations)**.** *The following considerations are crucial.*

- *We have introduced the continuous functional calculus that is the unique function*

$$\phi : C(\sigma(A)) \to \mathcal{B}(H)$$

  that is a $*$-homomorphism.
- *We now extend the calculus ϕ to a calculus ψ defined for function that are bounded and measurable $f : \mathbb{R} \to \mathbb{C}$ that we will denote by $B(\mathbb{R})$ with the norm*

$$\|f\|_\infty := \sup_{t\in\mathbb{R}} |f(t)|.$$

- *The calculus ψ is an extension of the calculus ϕ since it will have the property that*

$$\psi(f) = \phi(f)$$

  for all $f \in C(\sigma(A))$.

Theorem 2.2 (Spectral Theorem as functional calculus (version 1))**.** *Let A be a self-adjoint operator in $\mathcal{B}(H)$, where H is a complex separable Hilbert space H. Then there exists a unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$ that has the following properties:*

(A) *(Extension) For $A \in \mathcal{B}(H)$ and $f \in C(\sigma(A))$ then $\psi(f) = \phi(f)$ where ψ is the extended functional calculus and ϕ is the continuous functional calculus.*
(B) $\|\psi(g)\|_{\mathcal{B}(H)} \leq \|g\|_\infty, \quad \forall g \in B(\mathbb{R})$.
(C) *ψ is a $*$-homomorphism.*
(D) *If g_n and $g \in B(\mathbb{R})$ with $g_n \to g$ for $t \in \mathbb{R}$ and $\sup_{n\in\mathbb{N}} |g_n| < 0$, then*

$$\psi(g_n)x \to \psi(g)x, \quad \forall x \in H.$$

The proof is split into steps formulated in the following lemmas.

Lemma 2.3. *Let A be a self-adjoint operator in $\mathcal{B}(H)$, where H is a complex separable Hilbert space H. Then there exists a unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$ that has the following properties:*

(A) *(Extension) For $A \in \mathcal{B}(H)$ and $f \in C(\sigma(A))$ then $\psi(f) = \phi(f)$ where ψ is the extended functional calculus and ϕ is the continuous functional calculus.*

Proof. We consider the facts for the existence of the unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$

- We consider the unique function $\phi : C(\sigma(A)) \to \mathcal{B}(H)$ that is a $*$-homomorphism (the continuous functional calculus) and we define the linear functional:
$$\Lambda_{x,y}(f) := (\phi(f)x, y), \qquad \forall f \in C(\sigma(A))$$
- We show that $\Lambda_{x,y}(f)$ is bounded:
$$|\Lambda_{x,y}(f)| = |(\phi(f)x, y)| \leq \|\phi(f)x\|_H \|y\|_H \leq \|\phi(f)\|_{\mathcal{B}(H)} \|x\|_H \|y\|_H$$
- Since for the continuous functional calculus we have the estimate
$$\|\phi(f)\|_{\mathcal{B}(H)} = \sup_{\lambda \in \sigma(A)} |f(t)| = \|f\|_\infty$$
we get
$$|\Lambda_{x,y}(f)| \leq (\|x\|_H \|y\|_H) \, \|f\|_\infty.$$
- This means that for x and y fixed in H the functional
$$\Lambda_{x,y}(f) : C(\sigma(A)) \to \mathbb{C}$$
is bounded and the norm
$$\|\Lambda_{x,y}\| := \sup_{f \neq 0} \frac{|\Lambda_{x,y}(f)|}{\|f\|_\infty}$$
satisfy
$$\|\Lambda_{x,y}\| \leq \|x\|_H \|y\|_H.$$
- For the Riesz-Markov Theorem, there exists a unique function of bounded variation
$$F_{x,y} : \sigma(A) \to \mathbb{C}$$
such that
$$\Lambda_{x,y}(f) := (\phi(f)x, y) = \int_{\sigma(A)} f(t) \, dF_{x,y}(t), \quad \forall f \in C(\sigma(A)).$$

We extend the formula
$$\Lambda_{x,y}(f) := (\phi(f)x, y) = \int_{\sigma(A)} f(t) \, dF_{x,y}(t), \quad \forall f \in C(\sigma(A))$$
to functions in $g \in B(\sigma(A))$. We now have to prove that map
$$\Lambda_{x,y}(g) := \int_{\sigma(A)} g(t) \, dF_{x,y}(t)$$

is bounded and hermitian form in $x, y \in H$ for all $g \in B(\sigma(A))$. [1] The form $\Lambda_{x,y}(g)$ is bounded because

$$|\Lambda_{x,y}(g)| = |\int_{\sigma(A)} g(t) dF_{x,y}(t)| \leq \|x\|\|y\|\|g\|_\infty, \quad g \in B([a,b]).$$

To show that is hermitian we have to verify the properties of the sesquilinear form. [2]

We only show that $\Lambda_{x+y,z}(g) = \Lambda_{x,z}(g) + \Lambda_{y,z}(g)$ the other properties follows similarly. We consider for the moment g continuous and we take $f \in C(\sigma(A))$

$$\begin{aligned}\int_{\sigma(A)} f(t) dF_{x+y,z}(t) &= (\phi(f)x, z) + (\phi(f)y, z) \\ &= \int_{\sigma(A)} f(t) dF_{x,z}(t) + \int_{\sigma(A)} f(t) dF_{y,z}(t) \\ &= \int_{\sigma(A)} f(t) \left(dF_{x,z}(t) + dF_{y,z}(t)\right), \quad \forall f \in C(\sigma(A)).\end{aligned}$$

So we have

$$\int_{\sigma(A)} f(t) dF_{x+y,z}(t) = \int_{\sigma(A)} f(t) \Big(dF_{x,z}(t) + dF_{y,z}(t)\Big), \quad \forall f \in C(\sigma(A)).$$

From the relation

$$\int_{\sigma(A)} f(t) dF_{x+y,z}(t) = \int_{\sigma(A)} f(t) \Big(dF_{x,z}(t) + dF_{y,z}(t)\Big), \quad \forall f \in C(\sigma(A)).$$

and by the consequence of the Riesz-Markov theorem [3] We obtain

$$F_{x+y,z}(t) = F_{x,z}(t) + F_{y,z}(t)$$

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[1] The fact that is bounded is due to the fact that the estimate

$$\Big|\Lambda(f)\Big| = \Big|\int_a^b f(t)\, dF(t)\Big| \leq \|\Lambda\| \sup_{t\in[a,b]} |f(t)|, \quad \forall f \in C([a,b]),$$

still holds for the larger class of functions $f \in B([a,b])$, where $\|\Lambda\| \leq \|x\|\|y\|$.

[2] Let H be a Hilbert space. A hermitian form (or sesquilinear form) is a map

$$\Lambda_{\cdot,\cdot}(g) : H \times H \to \mathbb{C}$$

such that, for all x,y,$z \in H$ and $\lambda \in \mathbb{C}$ we have

- $\Lambda_{x+y,z}(g) = \Lambda_{x,z}(g) + \Lambda_{y,z}(g)$
- $\Lambda_{x,z+w}(g) = \Lambda_{x,z}(g) + \Lambda_{x,w}(g)$
- $\Lambda_{\lambda x,y}(g) = \lambda \Lambda_{x,y}(g)$
- $\Lambda_{x,\lambda y}(g) = \overline{\lambda} \Lambda_{x,y}(g)$

[3] Let F_1 and F_2 be functions of bounded variation such that

$$\int_a^b f(t)\, dF_1(t) = \int_a^b f(t)\, dF_2(t), \quad \forall f \in C([a,b]).$$

Then $F_1(t) = F_2(t), \quad t \in [a,b]$. This fact is a consequence of the uniqueness if the function F as stated in the representation theorem.

now integrating $g \in B$ with respect to the function $F_{x+y,z}(t)$ of bounded variation we get

$$\int_{\sigma(A)} g(t)dF_{x+y,z}(t) = \int_{\sigma(A)} g(t)dF_{x,z}(t) + \int_{\sigma(A)} g(t)dF_{y,z}(t), \quad \forall g \in B(\sigma(A)).$$

Remark 2.4 (Summarizing about the operator $\psi(g)$)**.** *The form*

$$\Lambda_{x,y}(g) := \int_{\sigma(A)} g(t)dF_{x,y}(t)$$

is bounded and hermitian form in $x, y \in H$ for all $g \in B(\sigma(A))$ by the representation theorem of sesquilinear forms in Hilbert spaces [4] *there exists a unique bounded operator on H, that we denote by $\psi(g)$, such that*

$$\Lambda_{x,y}(g) = \int_{\sigma(A)} g(t)dF_{x,y}(t) = (\psi(g)x, y), \quad \forall x, y \in H, \quad g \in B(\mathbb{R}).$$

For $A \in \mathcal{B}(H)$ and $f \in C(\sigma(A))$ then

$$\psi(f) = \phi(f)$$

where ψ is the extended functional calculus and ϕ is the continuous functional calculus.

This is because if $f \in B(\mathbb{R})$ is continuous on a interval that contains $\sigma(A)$ it is

$$(\psi(f)x, y) = \int_a^b f(t)dF_{x,y}(t) = (\phi(f)x, y), \quad \forall x, y \in H$$

so

$$(\psi(f)x, y) = (\phi(f)x, y), \quad \forall x, y \in H \quad \Rightarrow \quad \psi(f) = \phi(f).$$

□

Lemma 2.5 (Property (B))**.** *Let A be a self-adjoint operator in $\mathcal{B}(H)$, where H is a complex separable Hilbert space H. Then there exists a unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$ that has the following property:*

(B) $\|\psi(g)\|_{\mathcal{B}(H)} \leq \|g\|_\infty, \quad \forall g \in B(\mathbb{R})$.

Proof. It follows from the estimate that we have obtained for the boundedness of the form $\Lambda_{x,y}(f)$ is bounded

$$|\Lambda_{x,y}(g)| = |\int_{\sigma(A)} g(t)dF_{x,y}(t)| \leq \|x\|\|y\|\|g\|_\infty, \quad g \in B([a, b]).$$

□

[4] Let $F : H \times H \to \mathbb{C}$ be a bounded sesquilinear form ($|F(x, y)| \leq M\|x\|\|y\|$). Then there exists a unique operator $A \in \mathcal{B}(H)$ such that

$$F(x, y) = (Ax, y) \ \ \forall x, y \in H$$

with

$$\|A\| = M.$$

Lemma 2.6 (Property (C)). *Let A be a self-adjoint operator in $\mathcal{B}(H)$, where H is a complex separable Hilbert space H. Then there exists a unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$ that has the following property:*

(C) *ψ is a $*$-homomorphism.*

Proof. We have to prove the $*$-homomorphism properties, [5] since we use the fact that ψ is an extension of ϕ we will denote f_1 and f_2 the functions in C, while g_1 and g_2 are functions in $B(\mathbb{R})$. The condition $\psi(\lambda g_1 + \mu g_2) = \lambda\psi(g_1) + \mu\psi(g_2)$ follows by the linearity of the integral

$$\begin{aligned}(\psi(\lambda g_1 + \mu g_2)x, y) &= \int_{\sigma(A)} (\lambda g_1(t) + \mu g_2(t)) dF_{x,y}(t) \\ &= \lambda(\psi(g_1)x, y) + \mu(\psi(g_2)x, y), \quad \forall x, y \in H\end{aligned}$$

The condition $\psi(1) = I$ follows from the fact that $\phi(1) = \psi(1) = I$.

The condition $\psi(g_1 g_2) = \psi(g_1)\psi(g_2)$ is more delicate. It is based on the fact that we have proved already that the continuous functional calculus ϕ is a $*$-homomorphism, in particular $\phi(f_1 f_2) = \phi(f_1)\phi(f_2)$, $\forall f_1 f_2 \in C([a, b])$ so

$$(\phi(f_1 f_2)x, y) = (\phi(f_1)\phi(f_2)x, y) \quad \forall x, y \in H$$

Now observe that for the representation

$$(\phi(f_1)x, y) = \int_{\sigma(A)} f_1(t)\, dF_{x,y}(t), \quad \forall f_1 \in C(\sigma(A)).$$

we can view the term $\phi(f_2)x$ a an element in H and

$$(\phi(f_1)\phi(f_2)x, y) = \int_{\sigma(A)} f_1(t)\, dF_{\phi(f_2)x,y}(t), \quad \forall x, y \in H \quad \forall f_1, f_2 \in C([a, b]).$$

So the relation becomes ($\forall x, y \in H, \forall f_1, f_2 \in C([a, b])$.)

$$\int_{\sigma(A)} f_1(t) f_2(t)\, dF_{x,y}(t) = \int_{\sigma(A)} f_1(t)\, dF_{\phi(f_2)x,y}(t),$$

by the property of the Stieltjes

$$\int_{\sigma(A)} f_1(t) d\Big(\int_a^t f_2(\eta)\, dF_{x,y}(\eta)\Big) = \int_{\sigma(A)} f_1(t)\, dF_{\phi(f_2)x,y}(t),$$

and since it is true $\forall f_1 \in C([a, b])$ we get the relation

$$\int_a^t f_2(\eta)\, dF_{x,y}(\eta) = F_{\phi(f_2)x,y}(t), \quad t \in [a, b]$$

[5] Definition of $*$-homomorphism: The functional calculus $g \to \psi(f) = g(A)$, for bounded self-adjoint operators A on H, for every g_1, $g_2 \in B(\mathbb{R})$ and λ, $\mu \in \mathbb{C}$, satisfies the conditions

- $\psi(\lambda g_1 + \mu g_2) = \lambda\psi(g_1) + \mu\psi(g_2)$
- $\psi(g_1 g_2) = \psi(g_1)\psi(g_2)$
- $\psi(1) = I$
- $\psi(\overline{g_1}) = \psi(g_1)^*$

now we take $g_1 \in B(\mathbb{R})$ and we integrate with respect to the measure $F_{\phi(f_2)x,y}(t)$ for $f_2 \in C([a,b])$.

So for $g_1 \in B(\mathbb{R})$ and $F_{\phi(f_2)x,y}(t)$ for $f_2 \in C([a,b])$, we have a well-defined functional:

$$\int_{\sigma(A)} g_1(t)\, d\left(\int_a^t f_2(\eta)\, dF_{x,y}(\eta)\right) = \int_{\sigma(A)} g_1(t)\, dF_{\phi(f_2)x,y}(t),$$

that is,

$$\int_a^b g_1(t) f_2(t)\, dF_{x,y}(t) = \int_a^b g_1(t)\, dF_{\phi(f_2)x,y}(t),$$

which can be written as

$$(\psi(g_1 f_2)x, y) = (\psi(g_1)\phi(f_2)x, y), \quad \forall x, y \in H.$$

Now we repeat the argument to replace $\phi(f_2)$ by $\psi(g_2)$.

We now take the adjoint $\psi(g_1)^*$ of $\psi(g_1)$ in

$$(\psi(g_1 f_2)x, y) = (\psi(g_1)\phi(f_2)x, y), \forall x, y \in H.$$

and we get

$$(\psi(g_1 f_2)x, y) = (\psi(g_1)\phi(f_2)x, y) = (\phi(f_2)x, \psi(g_1)^* y), \forall x, y \in H.$$

Similarly, we can write

$$\int_a^b f_2(t) d\Big(\int_a^t g_1(\eta) dF_{x,y}(\eta)\Big) = \int_a^b f_2(t) dF_{x,\psi(g_1)^* y}$$

since this relation holds for all $f_2 \in C([a,b])$ we get

$$\int_a^t g_1(\eta) dF_{x,y}(\eta) = dF_{x,\psi(g_1)^* y}.$$

Reasoning as before we integrate a function $g_2 \in B(\mathbb{R})$ with respect to the measure $dF_{x,\psi(g_1)^* y}$

$$\int_a^b g_2(t) d\Big(\int_a^t g_1(\eta) dF_{x,y}(\eta)\Big) = \int_a^b g_2(t) dF_{x,\psi(g_1)^* y}$$

so we get

$$\int_a^b g_2(t) g_1(t) dF_{x,y}(t) = \int_a^b g_2(t) dF_{x,\psi(g_1)^* y}$$

and so the statement follows

$$(\psi(g_1 g_2)x, y) = (\psi(g_1)\psi(g_2)x, y), \forall x, y \in H \quad \text{and} \quad g_1, g_2 \in B(\mathbb{R}).$$

From the fact that $\phi(f)$ is a $*$-homomorphism we get

$$(\phi(f_1)x, y) = (x, \phi(f_1)^* y) = (x, \phi(\overline{f_1})y) = \overline{(\phi(\overline{f_1})y, x)}$$

this means

$$\int_a^b f_1(t) dF_{x,y}(t) = \overline{\int_a^b \overline{f_1(t)} dF_{x,y}(t)} = \int_a^b f_1(t) d\overline{F_{x,y}(t)}$$

Recall the conjugation of complex numbers

$$\overline{\overline{z} \quad w} = \overline{\overline{z}} \;\; \overline{w} = z \;\; \overline{w}$$

since the relation hold $f_1 \in C([a,b])$ it is

$$\overline{F_{y,x}(t)} = F_{x,y}(t)$$

as a consequence

$$\int_a^b g_1(t) dF_{x,y}(t) = \int_a^b g_1(t) d\overline{F_{y,x}(t)} = \overline{\int_a^b \overline{g_1(t)} dF_{y,x}(t)}$$

so

$$(\psi(g_1)x, y) = \overline{(\psi(\overline{g_1})y, x)} = (x, \psi(\overline{g_1})y)$$

that implies $\psi(g_1)^* = \psi(\overline{g_1})$.

□

Lemma 2.7 (Propery (D))**.** *Let A be a self-adjoint operator in $\mathcal{B}(H)$, where H is a complex separable Hilbert space H. Then there exists a unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$ that has the following property:*

(D) *If g_n and $g \in B(\mathbb{R})$ with $g_n \to g$ for $t \in \mathbb{R}$ and $\sup_{n\in\mathbb{N}} |g_n| < 0$, then*

$$\psi(g_n)x \to \psi(g)x, \quad \forall x \in H.$$

Proof. Let g_n and g satisfy the conditions If g_n and $g \in B(\mathbb{R})$ with $g_n \to g$ for $t \in \mathbb{R}$ and $\sup_{n\in\mathbb{N}} |g_n| < 0$. Since ψ is a $*$-homomorphism we have

$$\begin{aligned}
\|\psi(g)x - \psi(g_n)x\|^2 &= (\psi(g-g_n)x, \psi(g-g_n)x) \\
&= (\psi(g-g_n)^*\psi(g-g_n)x, x) \\
&= (\psi(\overline{g-g_n})\psi(g-g_n)x, x) \\
&= (\psi(\overline{(g-g_n)}(g-g_n)x, x) \\
&= (\psi(|g-g_n|^2 x, x) \\
&= \int_a^b |g-g_n|^2 \, dF_{x,x}
\end{aligned}$$

Passing to the limit under the integral we get the statement

$$\|\psi(g)x - \psi(g_n)x\|^2 = \int_a^b |g-g_n|^2 dF_{x,x} \to 0 \quad \text{as} \quad n \to \infty.$$

□

3. Appendix A: Preparatory results

3.1. **Riemann-Stieltjes Integral.** The Riemann-Stieltjes integral is a generalization of the concept of the definite integral, obtained by replacing the integration variable with an appropriate function.

Definition 3.1 (Riemann-Stieltjes Integral sums)**.** *Consider, for example, a bounded function $f(x)$ and a continuous function $g(x)$ defined on an interval $[a, b]$. Given a partition of the interval by a sequence of points $\{x_k\}$ with*

$$a = x_0 < x_1 < \cdots < x_n = b,$$

and denoting by m_k and M_k the lower and upper bounds of $f(x)$ in $[x_{k-1}, x_k]$, one constructs the lower integral sums:

$$S_{lower} = \sum_{k=1}^{n} m_k[g(x_k) - g(x_{k-1})],$$

and the upper integral sums:

$$S_{upper} = \sum_{k=1}^{n} M_k[g(x_k) - g(x_{k-1})].$$

Remark 3.2. *The definition can be extended to function g in a larger class with respect to the continuous functions space.*

Definition 3.3 (Riemann-Stieltjes Integrability)**.** *If, as the maximum length δ of the intervals $[x_{k-1}, x_k]$ tends to zero, the lower and upper sums converge to the same limit J, then the function $f(x)$ is said to be Riemann-Stieltjes integrable with respect to the function $g(x)$, and this is written as:*

$$\int_a^b f(x)\, dg(x) = J.$$

The definition can then be extended to the interval $(-\infty, +\infty)$.

Lemma 3.4 (Riemann-Stieltjes Integral for $g(x)$ of Class C^1)**.** *If $g(x)$ is of class C^1, the integral coincides with:*

$$\int_a^b f(x)\, dg(x) = \int_a^b f(x)g'(x)\, dx,$$

and in particular, if $g(x) = x$, it reduces to the Riemann integral.

Example 3.5 (The delta representation by the Heaviside Function Example)**.** *In the general case, this results in a generalization that includes measures on $\mathbb{R}$. For example, if $g(x) = Y(x)$ (with Y being the Heaviside function defined as follows:*

$$Y(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

and $f(x)$ is continuous at the origin, we have:

$$\int_{-\infty}^{+\infty} f(x)\, dY(x) = f(0),$$

which means that the integral represents the Dirac delta distribution $\delta(x)$.

3.2. **Functions of bounded variation.**

Definition 3.6 (Total variation of a function)**.** *Let $I \subset \mathbb{R}$ be an interval and consider the collection Π of ordered $(N+1)$-tuple of points $a_1 < a_2 < \cdots < a_{N+1} \in I$, where N is an arbitrary natural number. The total variation of a function $f : I \to \mathbb{R}$ is given by:*

$$TV(f) := \sup \left\{ \sum_{i=1}^{N} |f(a_{i+1}) - f(a_i)| : (a_1, \ldots, a_{N+1}) \in \Pi \right\}.$$

Definition 3.7 (Classical definition (but careful!!))**.** *Let $I \subset \mathbb{R}$ be an interval. A function $f : I \to \mathbb{R}$ is said to have bounded variation if its total variation is bounded.*

A fundamental characterization of functions of bounded variation of one variable is due to Jordan.

Theorem 3.8 (General Properties)**.** *Let $I \subset \mathbb{R}$ be an interval. A function $f : I \to \mathbb{R}$ has bounded variation if and only if it can be written as the difference of two bounded, nondecreasing functions.*

Remark 3.9 (canonical representation in terms of difference of nondecreasing functions)**.** *Indeed, it is possible to find a canonical representation of any function of bounded variation as the difference of nondecreasing functions. (See Theorem 4 of Section 5.2 in [Ro]).*

Theorem 3.10 (Jordan Decomposition (See Theorem 3 of Section 5.2 in [Ro)**.** *)] If $f : [a, b] \to \mathbb{R}$ is a function of bounded variation, then there is a pair of increasing functions f^+ and f^- such that:*

$$f = f^+ - f^- \quad \text{and} \quad TV(f) = f^+(b) - f^+(a) + f^-(b) - f^-(a).$$

The pair is unique up to the addition of a constant, i.e., if g^+ and g^- is a second pair with the same property, then $g^+ - g^- = f^+ - f^- \equiv const.$

Theorem 3.11 (The right and left limits)**.** *If $f : I \to \mathbb{R}$ is a function of bounded variation, then:*

- *The right and left limits*

$$f(x^+) := \lim_{y \downarrow x} f(y), \quad f(x^-) := \lim_{y \uparrow x} f(y)$$

  exist at every point $x \in I$.
- *The set of points of discontinuity of f is at most countable.*

Remark 3.12 (**Warning**). *However, according to the definitions given above, it may happen that at a given point the right and left limits coincide, but nonetheless the function f is discontinuous. For instance, the function $f : \mathbb{R} \to \mathbb{R}$ given by:*

$$f(x) = \begin{cases} 1 & \text{if } x = 0, \\ 0 & \text{otherwise} \end{cases}$$

is a function of bounded variation.

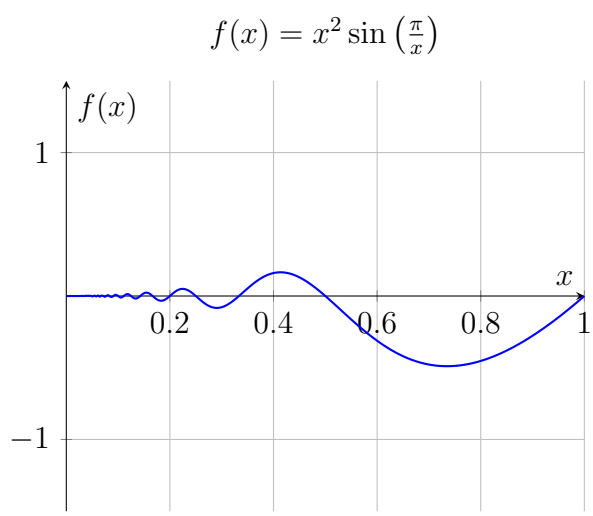
Remark 3.13 (Precise Representative)**.** *In order to avoid pathologies like in the* ***Warning remark****, it is customary to postulate some additional assumptions for functions of bounded variation. The suitable choice for us is*

- *Imposition of right (resp. left) continuity, i.e., at any point x, we impose:*

$$f(x) = f(x^+) \quad (\text{resp. } f(x) = f(x^-)).$$

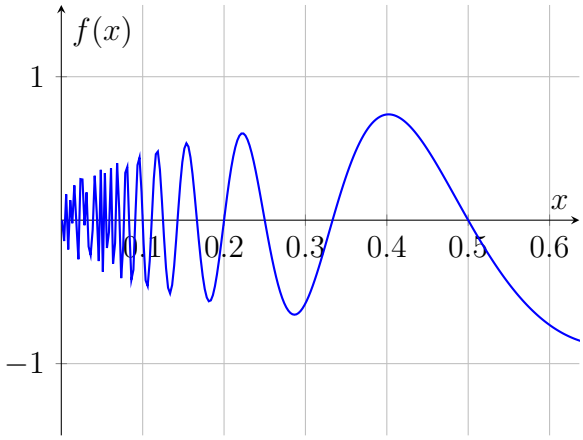
3.3. **Illustrations of functions in BV and not in BV.**

Remark 3.14. *Continuous function and in BV: $f(x) = x^2 \sin\left(\frac{\pi}{x}\right)$ and $f(0) = 0$*



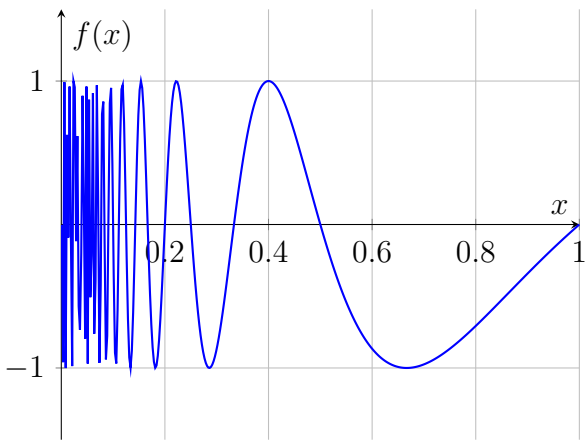
Remark 3.15. *It is continuous, but it is not BV: $f(x) = \sqrt[3]{x} \sin\left(\frac{1}{x}\right)$ and $f(0) = 0$,*

$f(x) = \sqrt[3]{x}\sin\left(\frac{\pi}{x}\right)$



Remark 3.16. *Not continuous function, and not BV:* $f(x) = \sin\left(\frac{\pi}{x}\right)$ *and* $f(0) = 0$

$f(x) = \sin\left(\frac{\pi}{x}\right)$



3.4. **Existence of the Riemann-Stieltjes Integral for functions of bounded variations.**

Theorem 3.17 (Existence Theorem for the Riemann-Stieltjes Integral)**.** *Let* $f : [a, b] \to \mathbb{R}$ *be a continuous function, and let* $g : [a, b] \to \mathbb{R}$ *be a function of bounded variation on* $[a, b]$*. Then, the Riemann-Stieltjes integral*

$$\int_a^b f(x)\, dg(x)$$

exists.

Moreover a remarkable fact is that

Theorem 3.18. *A curve is rectifiable if and only it is of bounded variation.*

Remark 3.19 (Explanation)**.** ***Key points:***

- *The function $f(x)$ is continuous, which ensures that it does not have any jumps or discontinuities. This property is essential for the integrability with respect to $g(x)$.*
- *The function $g(x)$ has bounded variation, meaning its total variation is finite. This condition is weaker than requiring $g(x)$ to be differentiable or continuous.*
- *The theorem guarantees that the Riemann-Stieltjes integral exists for this combination of $f(x)$ and $g(x)$.*

Remark 3.20. *The existence theorem for the Riemann-Stieltjes integral is important because it allows us to define integrals in cases where $g(x)$ is not necessarily differentiable, but only needs to have bounded variation. This includes a wide range of practical applications where the integrator function may be monotonic, stepwise, or oscillatory with controlled variation.*

Theorem 3.21 (Riesz Representation Theorem for Continuous Functions)**.** *Let $[a, b]$ be a closed interval in $\mathbb{R}$, and let $C([a, b])$ be the space of continuous functions on $[a, b]$.*

Then, for every bounded linear functional L on $C([a, b])$, there exists a unique function g of bounded variation on $[a, b]$ such that:

$$L(f) = \int_a^b f(x)\, dg(x)$$

for all $f \in C([a, b])$.

Moreover, given the bounded linear functional L its norm is:

$$\|L\| = \sup_{\|f\|_\infty \leq 1} |L(f)| = TV(g).$$